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CURVED-SURFACE BUTTON STRUCTURE

Abstract

A curved-surface button structure having a housing, a button arranged in the housing, a button assembly and a positioning sleeve. The housing has a through hole, a curved surface and an internal surface. The button has a driven part. The button assembly has a pressing plate, an installation seat and an extending column. The installation seat is arranged on the internal surface. The pressing plate is connected to the installation seat. The extending column is extended from the pressing plate toward the button. The positioning sleeve is adapted to sheathe the extending column. The positioning sleeve is larger than the driven part. When the button assembly is pressed, the extending column abuts against the driven part, and the positioning sleeve surrounds the driven part to effectively position a movement range of the button assembly to prevent from getting stuck or slipping off.

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Background/Summary

BACKGROUND OF THE INVENTION

Technical Field

[0001] The technical field relates to a button structure, in particular to a curved-surface button structure applied on a curved surface.

Description of Related Art

[0002] With a vigorous development of electronic technology, various electronic devices have been provided in the market for various requirements of consumers. The electronic device generally has a housing for protecting electronic components therein so as to prevent the electronic components from damage caused by collisions with foreign objects, dust adhesion, or water vapor. In addition, many electronic devices are provided with buttons allowing users quickly and intuitively operate specific functions.

[0003] A button has a keycap and a pressing piece. The keycap is accommodated in a hole of the housing and can be pressed by the user. The pressing piece is disposed at a bottom of the keycap corresponds to a button in the electronic device, so that when the user presses the keycap, the pressing piece is driven to trigger the button at the same time to transmit a signal. In general, the button also has a fixed seat disposed on an internal surface of the housing and multiple elastic structures connected between the fixed seat and the keycap, so that the pressed keycap can be reset by elastic forces of the elastic structures. Alternatively, multiple limiting structures may be disposed on the internal surface of the housing to limit a pressed range of the keycap, thereby preventing the keycap from getting stuck caused by skew during a pressed process which may let the button unable to be reset.

[0004] However, when the housing of the electronic device has a curvature, the user may press the keycap in an angle, in an area or at a position which leads to a shift of stress between the button and the button, and the button may be skewed therefore. Furthermore, a larger pressing force may make the pressing piece to slip off over the button, and the pressed button may get stuck or unable to be reset by the elastic forces of the elastic structures. Also, the curvature of the housing may make a side of the button to slip off over strokes of the limiting structures when pressed, and the pressing piece may shift from the button and therefore fail to trigger the button because of skew of the button. Therefore, when the pressing piece is shifted, the elastic structures may be broken because of over pressing by the user.

[0005] In views of this, in order to solve the above disadvantage, the inventor studied related technology and provided a reasonable and effective solution in this disclosure.

SUMMARY OF THE INVENTION

[0006] A purpose of this disclosure is effectively positioning a movement range of an assembly to avoid skew, and preventing the button assembly from getting stuck or trigger failure cause by skew or slipping off.

[0007] The disclosure is directed to a curved-surface button structure having a housing, a button, a button assembly and a positioning sleeve. The housing has a first case and a second case assembled with each other. The first case has a through hole, a curved surface and an internal surface. The curved surface and the internal surface are disposed at two sides of the first case opposite to each other. The through hole is communicated between the curved surface and the internal surface. The button is arranged in the housing, the button has a base and a driven part. The driven part is pressably connected with the base to face the internal surface. The button assembly has a pressing plate, an installation seat and an extending column. The installation seat is arranged on the internal surface. The pressing plate is connected to the installation seat and disposed corresponding to the through hole. The extending column is extended from the pressing plate toward the button. The

positioning sleeve is adapted to sheathe an end of the extending column. An inner diameter of the positioning sleeve is larger than an outer diameter of the driven part. When the button assembly is free from forcing, the pressing plate is accommodated in the through hole, and a gap is defined between the extending column and the driven part. When the button assembly is pressed by an external force, the pressing plate is retracted into the through hole, the extending column abuts against the driven part, and the positioning sleeve surrounds the driven part.

[0008] One of the exemplary embodiments, an installation distance is defined between an end surface of the positioning sleeve away from the first case and an end surface of the extending column away from the first case, and the installation distance is greater than or equal to 0.5 mm and less than or equal to 3 mm.

[0009] One of the exemplary embodiments, the installation distance is greater than or equal to 1.62 mm and less than or equal to 1.7 mm.

[0010] One of the exemplary embodiments, a distance between the positioning sleeve and the base is larger than the gap when the button assembly is free from forcing.

[0011] One of the exemplary embodiments, the positioning sleeve is free from contacting the base when the extending column abuts against the driven part.

[0012] One of the exemplary embodiments, the first case has a plurality of limiting pieces, each limiting piece is extended from the internal surface toward the second case, the limiting pieces surrounds the pressing plate to limit a movement range of the pressing plate.

[0013] One of the exemplary embodiments, the extending column is a column with a cross-shaped cross section, the extending column is tapered from the pressing plate toward the button, and the positioning sleeve is in a hollow cylinder shape.

[0014] One of the exemplary embodiments, the installation seat has a frame and a pair of elastic arms, the frame surrounds the pressing plate, each elastic arm has two ends respectively connected with the pressing plate and the frame.

[0015] One of the exemplary embodiments, each of the elastic arm is in an S shape, a wavy shape, a zigzag shape or a lightning shape.

[0016] One of the exemplary embodiments, the first case has a plurality of hooks, each hook is extended from the internal surface to the second case to snap the frame.

[0017] One of the exemplary embodiments, an inner diameter of the positioning sleeve is less than an inner diameter of the base.

[0018] One of the exemplary embodiments, the extending column and the positioning sleeve are formed as one piece.

[0019] One of the exemplary embodiments, when the button assembly is free from forcing, a spaced distance is defined between the positioning sleeve and the base, and the spaced distance is larger than the gap.

[0020] The curved-surface button structure of this disclosure is provided with a positioning sleeve at the end of the extending column, and the inner diameter of the positioning sleeve is larger than the outer diameter of the driven part. Moreover, the inner diameter of the positioning sleeve is smaller than the inner diameter of the base, so that the button assembly disposed on the curved surface is positioned by the positioning sleeve surrounding the driven part when pressed by an external force, thereby effectively positioning a movement range of the button assembly to avoid shifting from the button which is caused by skew. Furthermore, the button assembly is prevented from getting stuck of trigger failure caused by shifting or slipping off.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] The features of the disclosure believed to be novel are set forth with particularity in the

appended claims. The disclosure itself, however, may be best understood by reference to the following detailed description of the disclosure, which describes a number of exemplary embodiments of the disclosure, taken in conjunction with the accompanying drawings, in which:

[0022] FIG. **1** is an exploded view according to this disclosure.

[0023] FIG. **2** is a partial exploded view according to this disclosure.

[0024] FIG. **3** is a partially perspective view illustrating a first case and a button assembly according to this disclosure.

[0025] FIG. **4** is a partially perspective view according to this disclosure.

[0026] FIG. **5** is a partially exploded view according to this disclosure.

[0027] FIG. **6** is a partially exploded view in an initial status according to this disclosure.

[0028] FIG. **7** is a partially exploded view in a normally pressed status according to this disclosure.

[0029] FIG. **8** is a partially exploded view in an obliquely pressed status according to this disclosure.

DETAILED DESCRIPTION

[0030] The technical contents of this disclosure will become apparent with the detailed description of embodiments accompanied with the illustration of related drawings as follows. It is intended that the embodiments and drawings disclosed herein are to be considered illustrative rather than restrictive.

[0031] It should be understood that the orientations or positional relationships in this disclosure which are indicated by the terms such as “front side”, “rear side”, “left side”, “right side”, “front end”, “rear end”, “end”, “vertical”, “horizontal”, “vertical”, “top” and “bottom” are based on the orientations or positional relationships as shown in the drawings. These are only used for describing this disclosure and simplifying the description rather than indicating or implying that the device or element have a specific orientation or be constructed and operated in a specific orientation, and it should not be considered as limitations of the scopes of this disclosure.

[0032] The terms used herein such as “first,” “second,” “third,” “fourth,” and “fifth” for describing various elements, components, regions, levels or portions, these elements, components, regions, levels or portions should not be limited by these terms. These terms are only used for distinguishing one element, component, area, levels or section from another. Unless a sequence or order is clearly indicated in the context, terms such as “first,” “second,” “third,” “fourth” and “fifth” used herein do not imply the sequence or order.

[0033] The terms used herein without additional definition such as “substantially” and “approximately” are used to describe and illustrate small changes. When used in an event or situation, the term may include the precise moment at which the event or situation occurs, and a close approximation to moment the event or situation occurs. For example, when combined with a numerical value, the term may include a range of variation less than or equal to $\pm 10\%$ of the numerical value, such as less than or equal to $\pm 5\%$, less than or equal to $\pm 4\%$, less than or equal to $\pm 3\%$, less than or equal to $\pm 2\%$, less than or equal to $\pm 1\%$, less than or equal to $\pm 0.5\%$, less than or equal to $\pm 0.1\%$, or less than or equal to $\pm 0.05\%$.

[0034] Detailed descriptions and technical contents of this disclosure is described in the flowing paragraph with reference to the drawings. However, the drawings are attached only for illustration and are not intended to limit this disclosure.

[0035] This disclosure is directed to a curved-surface button structure as shown in FIGS. **1** to **6** which has a housing **10**, at least a button **20**, at least a button assembly **30** and at least a positioning sleeve **40**.

[0036] The housing **10** has a first case **11** and a second case **12** assembled with each other. According to this embodiment, the first case **11** is assembled on a side of the second case **12**, but scopes of this disclosure should not be limited to the embodiment. For example, the first case **11** may be assembled on a top or a bottom of the second case **12**. first case **11** has through hole **111**, a curved surface **112** and an internal surface **113**. Specifically, the curved surface **112** and the internal

surface **113** are disposed at two sides of the first case **11** opposite to each other, namely that the curved surface **112** is disposed on an external surface of the first case **11**. The internal surface **113** may be a flat plane, an inclined surface, an arc surface or a curved surface, but scopes of this disclosure should not be limited to the embodiment. The through hole **111** is communicated between the curved surface **112** and internal surface **113** to penetrate the first case **11**. A number of the through holes **111** is arranged corresponding to a required number of the button assembly **30**. According to this embodiment, two through holes **111** are provided, but scopes of this disclosure should not be limited to the embodiment. Additional button assemblies **30** and additional through holes **111** may be arranged according to various requirements. For example, one of the through holes **111** is described following for illustration.

[0037] According to this embodiment, two buttons **20** is arranged corresponding to two button assemblies **30**, but scopes of this disclosure should not be limited to the embodiment. For example, one of the buttons **20** is described following for illustration. The button **20** is disposed in the housing **10**. Specifically, the curved-surface button structure according to this disclosure further has a circuit board **50** the circuit board **50** is arranged in the housing **10** and fixed with at least one of the first case **11** and the second case **12**, and the button **20** is fixed on the circuit board **50** to be electrically connected with the circuit board **50**. The button **20** has a base **21** and a driven part **22**. The base **21** is fixed on the circuit board **50**. The driven part **22** is pressably connected to the base **21** to face the internal surface **113** of the first case **11**. Accordingly, the driven part **22** is retracted into the base **21** when pressed so as to act a signal transmission to perform an operation of specific functions.

[0038] According to this embodiment, two button assemblies **30** are provided, but scopes of this disclosure should not be limited to the embodiment. For example, one of the button assemblies **30** is described following for illustration. The button assembly **30** has a pressing plate **31**, an installation seat **32** and an extending column **33**. The installation seat **32** is arranged on the internal surface **113** of the first case **11**, An arrangement of the installation seat **32** is illustrated in later paragraph and not detailly described here. The pressing plate **31** is connected to the installation seat **32** and disposed corresponding to the through hole **111**. The extending column **33** is extended from the pressing plate **31** toward the button **20**. According to this embodiment, the extending column **33** is a column with a with a cross-shaped cross section, and the extending column **33** is an inverted cone which is tapered from the pressing plate **31** toward the button **20**, but the scoped of this disclosure should not be limited to the embodiment.

[0039] According to this embodiment, two positioning sleeves **40** are provided corresponding to the button assemblies **30**, but scopes of this disclosure should not be limited to the embodiment. For example, one of positioning sleeves **40** is described following for illustration. The positioning sleeve **40** is in a hollow cylinder shape. According to this embodiment, the positioning sleeve **40** is made of Polycarbonate (PC), so that it is easy for processing and controlling dimension, and has a rigidity which is not too soft, but scopes of this disclosure should not be limited to the embodiment. The positioning sleeve **40** sleeves an end of the extending column **33**. Specifically, according to this embodiment as shown in FIG. 5, the extending column **33** is tapered from the pressing plate **31** toward the button **20**, so that an internal surface of the positioning sleeve **40** can be fastened with an external surface on the end of the extending column **33**, namely that the positioning sleeve **40** has an end toward the pressing plate **31** which is fastened with the extending column **33**. Of course, the positioning sleeve **40** and the extending column **33** may be formed as one piece. Alternatively, the positioning sleeve **40** and the extending column **33** may be combined by bonding, buckling, welding or locking. In addition, an inner diameter of the positioning sleeve **40** is larger than the outer diameter of the driven part **22**, and the inner diameter of the positioning sleeve **40** is less than an inner diameter of the base **21**.

[0040] Referring to FIG. 6, accordingly, when the button assembly **30** is free from forcing, the pressing plate **31** is accommodated in the through hole **111** (in another embodiment, the pressing

plate **31** may partially protrudes from the curved surface **112** of the first case **11**), and a gap **d1** is defined between the end of the extending column **33** and a side of the driven part **22** away from the base **21**. Further refer to FIGS. **7** and **8**, when the button assembly **30** is pressed by an external force, the pressing plate **31** is retracted into the through hole **111** of the first case **11**, thereby driving the extending column **33** to abut against the driven part **22**. The positioning sleeve **40** surrounds the driven part **22** so as to be positioned and prevent the button assembly **30** from shifting from the button **20** caused by over skew, so that the button assembly **30** is further avoid from getting stuck or trigger failure caused by shifting or slipping off, and the extending column **33** can withstand greater pressing forces (approximately 3 kgf) in normal (see FIG. **7**), oblique (see FIG. **8**) or lateral directions without slipping off over the button **20**.

[0041] Furthermore, refer to FIGS. **5** and **6**, an installation distance **d2** is defined between an end surface of the positioning sleeve **40** away from the first case **11** and an end surface on the extending column **33** away from the first case **11**. The installation distance **d2** is greater than or equal to 0.5 mm and less than or equal to 3 mm. According to this embodiment, the installation distance **d2** is greater than or equal to 1.62 mm and less than or equal to 1.7 mm. A spaced distance **d3** is defined between the positioning sleeve **40** and the base **21**, and the spaced distance **d3** is larger than the gap **d1**. Accordingly, when the button assembly **30** is free from forcing namely in an initial state, the positioning sleeve **40** is ensured to surround a periphery on a top of the driven part **22**. When the button assembly **30** is in a pressed state namely the extending column **33** contacts the driven part **22**, an end surface of the positioning sleeve **40** away from the first case **11** is free from contacting the base **21** to avoid from interference. It is worth noting that during an assembly process in manufacturing, the installation distance **d2** between each positioning sleeve **40** and the extending column **33** correspondingly fixed therewith is stably and accurately controlled within the required range by tools, semi-automation machines or automation machines.

[0042] Further referring FIGS. **2** and **3**, the first case **11** also has a plurality of limiting pieces **114**. Each limiting piece **114** is extended from the inner surface **113** toward the second case **12**. The limiting pieces **114** surrounds a periphery of the pressing plate **31** to limit a movement range of the pressing plate **31**. According to this embodiment, four limiting pieces **114** are provided and arranged at equal angles, but scopes of this disclosure should not be limited to the embodiment. For example, two three, or more than five limiting pieces **114** may be provided.

[0043] Please refer to FIGS. **1** to **6**. The installation seat **32** has a frame **321** and at least a pair of elastic arms **322**, and the first case **11** also has a plurality of hooks **115**. The frame **321** surrounds the pressing plate **31**. Each hook **115** is extended from the internal surface **113** of the first case **11** toward the second case **12**, and the hooks **115** buckle to fix the frame **321**. Of course, the fixing form of the installation seat **32** and the first case **11** is not limited to be fix by the approach mentioned above. For example, the installation seat **32** may be fixed on the internal surface **113** of the first case **11** by clamping, bonding, welding, locking, etc. Each elastic arm **322** has two ends connected to the pressing plate **31** and the frame **321** respectively. According to this embodiment, each elastic arm **322** is substantially S-shaped, thereby having a required structural strength and a good elastic deformability, but scopes of this disclosure should not be limited to the embodiment. For example, each elastic arm **322** may be a wavy shape, a zigzag shape or a lightning shape, etc. In addition, in this embodiment, each pressing plate **31** is configured with a pair of elastic arms **322**, but scopes of this disclosure should not be limited to the embodiment. For example, each pressing plate **31** may be configured with greater than or equal to three elastic arms **322**.

[0044] The curved-surface button structure of this disclosure is provided with a positioning sleeve **40** at the end of the extending column **33**, and the inner diameter of the positioning sleeve **40** is larger than the outer diameter of the driven part **22**. Moreover, the inner diameter of the positioning sleeve **40** is smaller than the inner diameter of the base **21**, so that the button assembly **30** disposed on the curved surface **112** is positioned by the positioning sleeve **40** surrounding the driven part **22** when pressed by an external force, thereby effectively positioning a movement range of the button

assembly **30** to avoid shifting from the button **20** which is caused by skew. Furthermore, the button assembly **30** is prevented from getting stuck or trigger failure caused by shifting or slipping off. Also, each elastic arm **322** is prevented from broken caused by over pressing by a user, and the extending column **33** can withstand greater pressing forces in normal, oblique or lateral directions without slipping off over the button **20**.

[0045] Accordingly, this disclosure is believed to be applicable in industry, novel and inventive. Although this disclosure has been described with reference to the foregoing embodiment, it will be understood that the disclosure is not limited to the details thereof. Various equivalent variations and modifications can still occur to those skilled in this art in view of the teachings of this disclosure. Thus, all such variations and equivalent modifications are also embraced within the scope of this disclosure as defined in the appended claims.

Claims

1. A curved-surface button structure, comprising: a housing, comprising a first case and a second case assembled with each other, the first case comprising a through hole, a curved surface and an internal surface, the curved surface and the internal surface arranged at two sides of the first case opposite to each other, the through hole communicating between the curved surface and the internal surface; a button, arranged in the housing, comprising a base and a driven part, the driven part pressably connected with the base to face the internal surface; a button assembly, comprising a pressing plate, an installation seat and an extending column, the installation seat arranged in the internal surface, the pressing plate connected with the installation seat corresponding to the through hole, the extending column extended from the pressing plate toward the button; and a positioning sleeve, adapted to sheathe an end of the extending column, and an inner diameter of the positioning sleeve being greater than an outer diameter of the driven part; wherein when the button assembly is free from forcing, the pressing plate is accommodated in the through hole, and a gap is defined between the extending column and the driven part; wherein when the button assembly is pressed by an external force, the pressing plate is retracted into the through hole, the extending column presses the driven part, and the positioning sleeve surrounds the driven part.
2. The curved-surface button structure according to claim 1, wherein an installation distance is defined between an end surface of the positioning sleeve away from the first case and an end surface of the extending column away from the first case, and the installation distance is greater than or equal to 0.5 mm and less than or equal to 3 mm.
3. The curved-surface button structure according to claim 2, wherein the installation distance is greater than or equal to 1.62 mm and less than or equal to 1.7 mm.
4. The curved-surface button structure according to claim 1, wherein a distance between the positioning sleeve and the base is larger than the gap, when the button assembly is free from forcing.
5. The curved-surface button structure according to claim 1, wherein the positioning sleeve is free from contacting the base, when the extending column abuts against the driven part.
6. The curved-surface button structure according to claim 1, wherein the first case comprises a plurality of limiting pieces, each limiting piece is extended from the internal surface toward the second case, and the limiting pieces surround the pressing plate to limit a movement range of the pressing plate.
7. The curved-surface button structure according to claim 1, wherein the extending column is a column with a cross-shaped cross section, the extending column is tapered from the pressing plate toward the button, and the positioning sleeve is in a hollow cylinder shape.
8. The curved-surface button structure according to claim 1, wherein the installation seat comprises a frame and a pair of elastic arms, the frame surrounds the pressing plate, each elastic arm comprises two ends respectively connected with the pressing plate and the frame.

9. The curved-surface button structure according to claim 8, wherein each of the elastic arm is in an S shape, a wavy shape, a zigzag shape or a lightning shape.

10. The curved-surface button structure according to claim 8, wherein the first case comprises a plurality of hooks, each hook is extended from the internal surface to the second case and snap the frame.

11. The curved-surface button structure according to claim 1, wherein an inner diameter of the positioning sleeve is less than an inner diameter of the base.

12. The curved-surface button structure according to claim 1, wherein the extending column and the positioning sleeve are formed as one piece.

13. The curved-surface button structure according to claim 1, wherein when the button assembly is free from forcing, a spaced distance is defined between the positioning sleeve and the base, and the spaced distance is larger than the gap.
